GENETICS CO., LTD.

Korean Holstein Sexed Dairy Embryos for the Republic of Uzbekistan

Company Profile and Proposed Framework for the Supply of 50,000 Embryos

Pre-read for the video conference with the
Ministry of Economy and Finance of the Republic of Uzbekistan

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Genetics Co., Ltd. · Korea Animal Hospital
Anseong, Gyeonggi-do, Republic of Korea

This document is a discussion draft. Figures marked as estimates or indicative are planning assumptions, not contractual commitments. Final volumes, schedule, prices and veterinary conditions will be set out in a separate commercial proposal and the agreed import health protocol.

1. Purpose of This Document

This document is shared ahead of our video conference to give the Ministry of Economy and Finance a concise view of three points:

- **Who we are** – Genetics Co., Ltd., a Korean livestock biotechnology company specialised in dairy embryo production and embryo transfer.
- **What we propose** – a phased programme to supply **50,000 Korean Holstein sexed (female) dairy embryos** to Uzbekistan, together with the technical services needed to turn embryos into pregnancies, heifers and milk.
- **What we would like to discuss** – programme scale and timing, financing and procurement channels, veterinary/import requirements, and the division of roles between the Uzbek and Korean sides.

2. Executive Summary

Key points

- **Product:** frozen, in-vitro produced (IVP) Holstein embryos fertilised with **sexed semen**, so that around 90% of calves are expected to be female (heifers).
- **Track record:** about **3,000 embryo transfers** and **4,000 embryos sold per year**; transfers of our sexed, frozen IVF embryos abroad achieved a **conception rate of about 60% over the last five years** (company field data).
- **Uzbekistan experience:** we have exported Korean Holstein sexed embryos to Uzbekistan for about three years in cooperation with Korea's **Rural Development Administration (RDA)**, and ran a smart-farming sensor pilot on about 50 cows.
- **Proposal:** 50,000 embryos delivered in **four annual phases (indicatively 5,000 → 10,000 → 15,000 → 20,000)**, starting with a validation phase and scaling up as results are confirmed.
- **Expected result (planning estimate):** about **30,000 pregnancies** and about **24,000 female calves**, i.e. a new generation of high-genetic-merit dairy heifers born on Uzbek farms.
- **Package, not just a product:** embryos + recipient selection + training of Uzbek veterinarians and technicians + digital follow-up + performance reporting to the Government.

Compared with importing live pregnant heifers, embryo import is designed to lower biosecurity risk, transport cost and animal losses, and uses Uzbekistan's own heifers as recipients, so investment and know-how stay in the country.

3. About Genetics Co., Ltd.

Item	Details
Company	Genetics Co., Ltd. (Republic of Korea), with affiliated Korea Animal Hospital – a veterinary hospital specialised in embryo transfer and reproduction
Established	2005
Location	Anseong, Gyeonggi-do – the centre of Korea's dairy industry
Core business	Dairy embryo production (IVF), embryo transfer, sexed embryos, genomic selection, reproduction consulting, recipient-cow programmes
Scale	About 3,000 embryo transfers per year; about 4,000 embryos sold per year
Veterinary team	Dr. Hyunjae Ha (CEO), Dr. Kwangtaek Lim, Dr. Donghwan Shin, Dr. Mincheol Kim, supported by laboratory researchers
International	Embryo export and on-site transfer support in Mongolia, Uzbekistan and Kyrgyzstan; training and field programmes in other countries

Item	Details
Related businesses	Song Young Shin Farm (A2A2 Jersey dairy and dairy processing); precision-livestock monitoring using rumen sensors (CowTalk AI); livestock environment technology (D2O)

Founder and CEO – Dr. Hyunjae Ha, DVM

- Doctor of Veterinary Medicine, Konkuk University (1999); Adjunct Professor, Konkuk University.
- More than 25 years in large-animal veterinary practice and 18+ years in embryo transfer and in-vitro fertilisation.
- Breeder of A2A2 Jersey cattle and operator of an organic dairy farm for over 15 years.
- First to introduce smaXtec rumen-sensor monitoring to Korean dairy farms.
- Hands-on experience of embryo export and local transfer programmes in Central Asia (Mongolia, Uzbekistan, Kyrgyzstan).

4. Why Korean Holstein Sexed Embryos

4.1 Korean Holstein genetics

- The Korean Holstein population has been improved for decades with top sires from North America and Europe and national milk recording.
- Korean herds are raised in intensive housing under hot summers and cold winters – conditions comparable in many respects to Uzbekistan's continental climate.
- Every embryo is fertilised with **sexed semen**, so the programme produces mainly **female calves** – the replacement heifers that dairy farms need.

4.2 Embryos compared with live heifer imports

Factor	Live pregnant heifer import	Embryo import (proposed)
Biosecurity	Live animals can carry infectious disease across borders; long quarantine required	Embryos handled to WOAH/IETS standards carry a much lower risk of disease transmission
Transport	Sea/air/road shipment of live animals; stress, losses and high freight cost	Thousands of embryos travel in liquid-nitrogen containers on a regular flight
Adaptation	Imported animals must adapt to new climate, feed and pathogens	Calves are born in Uzbekistan and develop local immunity from birth
Use of local assets	Local cattle are not used	Uzbek heifers are used as recipients; local farms, vets and technicians are involved
Know-how	Limited transfer of skills	Training of Uzbek veterinarians and technicians builds national capacity
Foreign currency	High cost per animal	Expected lower foreign-currency outlay per heifer born (to be quantified in the commercial proposal)

5. Our Production Technology and Quality Control

Step	What we do	Why it matters
Oocyte source	Oocytes collected from ovaries of Korean Holstein cows	Large-scale, cost-efficient embryo production from the Korean Holstein population
Fertilisation	In-vitro fertilisation with **sexed semen** from selected sires	About 90% female calves expected

Step	What we do	Why it matters
Individual culture	Each embryo is cultured **individually**, not in pooled batches	Every embryo keeps its dam and sire identity – full traceability for breeding records
Grading	Embryos graded by stage and quality following IETS guidance	Only transferable-grade embryos are frozen and exported
Freezing	**Slow freezing in ethylene glycol for direct transfer**	Thawed and transferred on the farm like a semen straw – simple for field technicians
Documentation	Embryo ID list, donor/sire records, health certificate per agreed protocol	Supports import approval and the national breeding database

Field results depend strongly on recipient management. Our ~60% conception rate abroad was achieved with well-managed recipients. For this programme our default recipient is a **maiden heifer of breeding age** with good body condition; parous cows are used only by exception with complete breeding and health records.

6. Proposed Framework for 50,000 Embryos

We propose a phased programme so that each phase is approved on the basis of measured results from the previous phase. Annual volumes below are **indicative** and will be confirmed in the commercial proposal.

Phase	Timing (indicative)	Embryos	Main activities
0. Preparation	3–6 months	–	Agree import health protocol and permits; select regions and recipient farms; train first group of Uzbek technicians; set up data reporting
1. Validation	Year 1	5,000	Transfers in selected clusters; measure conception, pregnancy and calving rates; refine logistics
2. Scale-up	Year 2	10,000	Expand to more regions; local technicians perform most transfers under supervision
3. Expansion	Year 3	15,000	Regional rollout; heifers from Phase 1 enter first lactation
4. Full scale	Year 4	20,000	National coverage; local team operates independently; long-term genetic improvement plan
Total	**~4 years**	**50,000**	

Transfers will be scheduled in spring and autumn windows to avoid peak summer heat, which reduces conception rates.

6.1 Expected outcomes (planning estimates)

Indicator	Assumption	Estimate
Embryos transferred	–	50,000
Pregnancies	~60% conception rate (company field data abroad)	~30,000
Calves born	~10% pregnancy loss to term	~27,000
Female calves (heifers)	~90% female from sexed semen	~24,000
Milk potential (illustrative)	If each heifer produces 6,000 kg in her first lactation	~145,000 tonnes of milk per lactation cycle

These are planning estimates for discussion only. Actual results depend on recipient quality, nutrition, heat management and farm conditions, and will be measured and reported during the programme.

6.2 Recipient requirement

A programme of this size requires about **50,000 prepared recipient heifers** over four years – roughly **60,000 heifers screened** if about 80% qualify on the day of transfer. Identifying and preparing recipients (age, weight, health, vaccination, synchronisation) is the most important factor for success and is a key topic for our meeting.

7. What the Package Includes and Division of Roles

Area	Genetics Co., Ltd. (Korea)	Uzbek side (to be agreed)
Embryos	Production, grading, freezing, export documents, delivery to Tashkent	Import permit, customs and veterinary clearance
Recipients	Selection criteria, synchronisation protocol, on-site screening support	Provision of recipient heifers and participating farms
Transfer	Transfers by our vets in early phases; supervision later	Local veterinarians and technicians assigned to the programme
Training	Training courses in Uzbekistan and in Korea (ET, thawing, pregnancy diagnosis, heifer rearing)	Selection of trainees; training venue
Follow-up	Pregnancy and calving tracking, digital records, optional rumen-sensor monitoring (CowTalk AI)	Farm data access; reporting focal point
Reporting	Regular performance reports to the Government	Government programme management

8. Financing and Cooperation Options for Discussion

We are open to the structure that best fits the Government's priorities. Options include:

- **Government procurement** of embryos and services, with distribution to farms under a national livestock development programme.
- **Farmer co-financing** through state subsidy or concessional credit, with Genetics Co. supplying embryos and services directly to participating farms.
- **Korea–Uzbekistan cooperation channels**, for example RDA (KOPIA) technical cooperation, KOICA, or Korea Eximbank's EDCF, subject to the eligibility rules and decisions of those institutions.
- **Long-term genetic improvement agreement**, combining embryo supply with training, data services and later expansion.

Prices, payment terms and delivery schedules will be presented in a separate commercial proposal after the scope is agreed.

9. Key Risks and How We Manage Them

Risk	Mitigation
Recipient availability and quality	Clear selection criteria; pre-screening; phased volumes linked to prepared recipients
Veterinary / import requirements	Early agreement of the import health protocol between the two veterinary authorities
Logistics (liquid nitrogen, flights)	Validated dry-shipper containers; regular direct flights Seoul–Tashkent; temperature records for every shipment
Heat stress	Transfers scheduled in spring and autumn; heat-management guidance for farms
Lower-than-expected results	Phase-gate review after each phase; corrective plan before the next volume is released
Production capacity	Annual volumes confirmed against our production plan before each phase

10. Proposed Agenda for the Video Conference

1. Introductions and objectives of both sides
2. Uzbekistan's dairy development targets (herd size, milk production, timeline)
3. Programme scale and phasing for 50,000 embryos
4. Recipient heifer supply and participating regions/farms
5. Financing and procurement structure
6. Veterinary and import requirements; responsible agencies
7. Next steps: working group, commercial proposal, site visit

Questions we would value the Ministry's view on

- Which national programme or budget line would this project belong to, and which agency would lead implementation?
- Preferred financing model: direct government procurement, farmer subsidy, international cooperation funds, or a combination?
- Which regions and farm types (large commercial farms, clusters, smallholders) should be prioritised?
- Expected timeline for the first transfers?

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